

HW-1

```
library(palmerpenguins)
```

Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

penguins, penguins_raw

line 7: loading in the library

```
dim(penguins)
```

```
[1] 344    8
```

```
str(penguins) # variable types present in the dataset.
```

```
tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
 $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm  : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g    : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex           : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year          : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

```
summary(penguins)
```

species	island	bill_length_mm	bill_depth_mm
Adelie :152	Biscoe :168	Min. :32.10	Min. :13.10
Chinstrap: 68	Dream :124	1st Qu.:39.23	1st Qu.:15.60
Gentoo :124	Torgersen: 52	Median :44.45	Median :17.30
		Mean :43.92	Mean :17.15
		3rd Qu.:48.50	3rd Qu.:18.70
		Max. :59.60	Max. :21.50
		NA's :2	NA's :2

flipper_length_mm	body_mass_g	sex	year
Min. :172.0	Min. :2700	female:165	Min. :2007
1st Qu.:190.0	1st Qu.:3550	male :168	1st Qu.:2007
Median :197.0	Median :4050	NA's : 11	Median :2008
Mean :200.9	Mean :4202		Mean :2008
3rd Qu.:213.0	3rd Qu.:4750		3rd Qu.:2009
Max. :231.0	Max. :6300		Max. :2009
NA's :2	NA's :2		

line 12: number of rows and columns in the dataset. there are 344 rows and 8 columns.

line 13: shows more information about all the variables and their corresponding set of values.

```
# 1.1: Extract and analyze bill lengths
bill_lengths <- penguins[["bill_length_mm"]]

mean(bill_lengths, na.rm = TRUE)
```

```
[1] 43.92193
```

```
median(bill_lengths, na.rm = TRUE)
```

```
[1] 44.45
```

```
sd(bill_lengths, na.rm = TRUE)
```

```
[1] 5.459584
```

```
min(bill_lengths, na.rm = TRUE)
```

```
[1] 32.1
```

```
max(bill_lengths, na.rm = TRUE)
```

```
[1] 59.6
```

line 21: Assigning the variable to the vector made from the “bill_length_mm” variable

lines 23-27: Finding all necessary metrics. “na.rm” makes the calculation ignore missing values in the calculation.

```
# 1.2: Logical subsetting
```

```
long_bills <- bill_lengths[which(bill_lengths > 45)]  
long_bills
```

```
[1] 46.0 45.8 45.6 46.1 50.0 48.7 50.0 47.6 46.5 45.4 46.7 46.8 49.0 45.5 48.4  
[16] 45.8 49.3 49.2 46.2 48.7 50.2 45.1 46.5 46.3 46.1 47.8 48.2 50.0 47.3 45.1  
[31] 59.6 49.1 48.4 48.7 49.6 45.3 49.6 50.5 45.5 50.5 45.2 46.6 48.5 45.1 50.1  
[46] 46.5 45.5 50.4 45.3 46.2 45.7 54.3 45.8 49.8 46.2 49.5 50.7 47.7 46.4 48.2  
[61] 46.5 46.4 48.6 47.5 51.1 45.2 45.2 49.1 52.5 47.4 50.0 50.8 51.3 47.5 52.1  
[76] 47.5 52.2 45.5 49.5 50.8 49.4 46.9 48.4 51.1 48.5 55.9 47.2 49.1 47.3 46.8  
[91] 53.4 48.1 50.5 49.8 51.5 46.2 55.1 48.8 47.2 46.8 50.4 45.2 49.9 46.5 50.0  
[106] 51.3 45.4 52.7 45.2 46.1 51.3 46.0 51.3 46.6 51.7 47.0 52.0 45.9 50.5 50.3  
[121] 58.0 46.4 49.2 48.5 50.6 46.7 52.0 50.5 49.5 46.4 52.8 54.2 51.0 49.7 47.5  
[136] 47.6 52.0 46.9 53.5 49.0 46.2 50.9 45.5 50.9 50.8 50.1 49.0 51.5 49.8 48.1  
[151] 51.4 45.7 50.7 52.2 45.2 49.3 50.2 45.6 51.9 46.8 45.7 55.8 49.6 50.8 50.2
```

Used the which function here because “na.rm” does not work in square brackets and I wanted to drop all missing values from the output.

```
# 2.1: Island Analysis
summary(penguins)
```

species	island	bill_length_mm	bill_depth_mm
Adelie :152	Biscoe :168	Min. :32.10	Min. :13.10
Chinstrap: 68	Dream :124	1st Qu.:39.23	1st Qu.:15.60
Gentoo :124	Torgersen: 52	Median :44.45	Median :17.30
		Mean :43.92	Mean :17.15
		3rd Qu.:48.50	3rd Qu.:18.70
		Max. :59.60	Max. :21.50
		NA's :2	NA's :2

flipper_length_mm	body_mass_g	sex	year
Min. :172.0	Min. :2700	female:165	Min. :2007
1st Qu.:190.0	1st Qu.:3550	male :168	1st Qu.:2007
Median :197.0	Median :4050	NA's : 11	Median :2008
Mean :200.9	Mean :4202		Mean :2008
3rd Qu.:213.0	3rd Qu.:4750		3rd Qu.:2009
Max. :231.0	Max. :6300		Max. :2009
NA's :2	NA's :2		

```
dream_bills <- penguins$bill_length_mm[penguins$island == "Dream"]
mean(dream_bills, na.rm = TRUE)
```

```
[1] 44.16774
```

line 44: extracts the penguins from Dream island and their bill lengths and saves them to a variable

line 45: calculates and returns the mean Dream island penguin bill length. ensured to get rid of missing values

```
# 2.2: Species Counting
sum(penguins$species == "Adelie")
```

```
[1] 152
```

Initially tried to simply assign all adelie penguins to a variable using the same framework as 2.1 above but it returned the word “Adelie” 152 times for all 152 Adelie penguins. Then, used the sum function to return a scalar.

```
# 2.3: Find the heaviest penguin
summary(penguins)
```

species	island	bill_length_mm	bill_depth_mm
Adelie :152	Biscoe :168	Min. :32.10	Min. :13.10
Chinstrap: 68	Dream :124	1st Qu.:39.23	1st Qu.:15.60
Gentoo :124	Torgersen: 52	Median :44.45	Median :17.30
		Mean :43.92	Mean :17.15
		3rd Qu.:48.50	3rd Qu.:18.70
		Max. :59.60	Max. :21.50
		NA's :2	NA's :2
flipper_length_mm	body_mass_g	sex	year
Min. :172.0	Min. :2700	female:165	Min. :2007
1st Qu.:190.0	1st Qu.:3550	male :168	1st Qu.:2007
Median :197.0	Median :4050	NA's : 11	Median :2008
Mean :200.9	Mean :4202		Mean :2008
3rd Qu.:213.0	3rd Qu.:4750		3rd Qu.:2009
Max. :231.0	Max. :6300		Max. :2009
NA's :2	NA's :2		

```
heaviest <- which.max(penguins$body_mass_g)
penguins$body_mass_g[heaviest]
```

```
[1] 6300
```

```
penguins$species[heaviest]
```

```
[1] Gentoo  
Levels: Adelie Chinstrap Gentoo
```

```
penguins$island[heaviest]
```

```
[1] Biscoe  
Levels: Biscoe Dream Torgersen
```

```
# 3.1: Create measurement matrix
```

```
penguins_complete <- penguins[complete.cases(penguins), ] # provided
```

```
penguin_measurements <- as.matrix(penguins_complete[, c("bill_length_mm", "bill_depth_mm", "flipper_length_mm", "body_mass_g")])  
penguin_measurements
```

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
[1,]	39.1	18.7	181	3750
[2,]	39.5	17.4	186	3800
[3,]	40.3	18.0	195	3250
[4,]	36.7	19.3	193	3450
[5,]	39.3	20.6	190	3650
[6,]	38.9	17.8	181	3625
[7,]	39.2	19.6	195	4675
[8,]	41.1	17.6	182	3200
[9,]	38.6	21.2	191	3800
[10,]	34.6	21.1	198	4400
[11,]	36.6	17.8	185	3700
[12,]	38.7	19.0	195	3450
[13,]	42.5	20.7	197	4500
[14,]	34.4	18.4	184	3325
[15,]	46.0	21.5	194	4200
[16,]	37.8	18.3	174	3400
[17,]	37.7	18.7	180	3600
[18,]	35.9	19.2	189	3800
[19,]	38.2	18.1	185	3950
[20,]	38.8	17.2	180	3800
[21,]	35.3	18.9	187	3800
[22,]	40.6	18.6	183	3550
[23,]	40.5	17.9	187	3200

[24,]	37.9	18.6	172	3150
[25,]	40.5	18.9	180	3950
[26,]	39.5	16.7	178	3250
[27,]	37.2	18.1	178	3900
[28,]	39.5	17.8	188	3300
[29,]	40.9	18.9	184	3900
[30,]	36.4	17.0	195	3325
[31,]	39.2	21.1	196	4150
[32,]	38.8	20.0	190	3950
[33,]	42.2	18.5	180	3550
[34,]	37.6	19.3	181	3300
[35,]	39.8	19.1	184	4650
[36,]	36.5	18.0	182	3150
[37,]	40.8	18.4	195	3900
[38,]	36.0	18.5	186	3100
[39,]	44.1	19.7	196	4400
[40,]	37.0	16.9	185	3000
[41,]	39.6	18.8	190	4600
[42,]	41.1	19.0	182	3425
[43,]	36.0	17.9	190	3450
[44,]	42.3	21.2	191	4150
[45,]	39.6	17.7	186	3500
[46,]	40.1	18.9	188	4300
[47,]	35.0	17.9	190	3450
[48,]	42.0	19.5	200	4050
[49,]	34.5	18.1	187	2900
[50,]	41.4	18.6	191	3700
[51,]	39.0	17.5	186	3550
[52,]	40.6	18.8	193	3800
[53,]	36.5	16.6	181	2850
[54,]	37.6	19.1	194	3750
[55,]	35.7	16.9	185	3150
[56,]	41.3	21.1	195	4400
[57,]	37.6	17.0	185	3600
[58,]	41.1	18.2	192	4050
[59,]	36.4	17.1	184	2850
[60,]	41.6	18.0	192	3950
[61,]	35.5	16.2	195	3350
[62,]	41.1	19.1	188	4100
[63,]	35.9	16.6	190	3050
[64,]	41.8	19.4	198	4450
[65,]	33.5	19.0	190	3600
[66,]	39.7	18.4	190	3900

[67,]	39.6	17.2	196	3550
[68,]	45.8	18.9	197	4150
[69,]	35.5	17.5	190	3700
[70,]	42.8	18.5	195	4250
[71,]	40.9	16.8	191	3700
[72,]	37.2	19.4	184	3900
[73,]	36.2	16.1	187	3550
[74,]	42.1	19.1	195	4000
[75,]	34.6	17.2	189	3200
[76,]	42.9	17.6	196	4700
[77,]	36.7	18.8	187	3800
[78,]	35.1	19.4	193	4200
[79,]	37.3	17.8	191	3350
[80,]	41.3	20.3	194	3550
[81,]	36.3	19.5	190	3800
[82,]	36.9	18.6	189	3500
[83,]	38.3	19.2	189	3950
[84,]	38.9	18.8	190	3600
[85,]	35.7	18.0	202	3550
[86,]	41.1	18.1	205	4300
[87,]	34.0	17.1	185	3400
[88,]	39.6	18.1	186	4450
[89,]	36.2	17.3	187	3300
[90,]	40.8	18.9	208	4300
[91,]	38.1	18.6	190	3700
[92,]	40.3	18.5	196	4350
[93,]	33.1	16.1	178	2900
[94,]	43.2	18.5	192	4100
[95,]	35.0	17.9	192	3725
[96,]	41.0	20.0	203	4725
[97,]	37.7	16.0	183	3075
[98,]	37.8	20.0	190	4250
[99,]	37.9	18.6	193	2925
[100,]	39.7	18.9	184	3550
[101,]	38.6	17.2	199	3750
[102,]	38.2	20.0	190	3900
[103,]	38.1	17.0	181	3175
[104,]	43.2	19.0	197	4775
[105,]	38.1	16.5	198	3825
[106,]	45.6	20.3	191	4600
[107,]	39.7	17.7	193	3200
[108,]	42.2	19.5	197	4275
[109,]	39.6	20.7	191	3900

[110,]	42.7	18.3	196	4075
[111,]	38.6	17.0	188	2900
[112,]	37.3	20.5	199	3775
[113,]	35.7	17.0	189	3350
[114,]	41.1	18.6	189	3325
[115,]	36.2	17.2	187	3150
[116,]	37.7	19.8	198	3500
[117,]	40.2	17.0	176	3450
[118,]	41.4	18.5	202	3875
[119,]	35.2	15.9	186	3050
[120,]	40.6	19.0	199	4000
[121,]	38.8	17.6	191	3275
[122,]	41.5	18.3	195	4300
[123,]	39.0	17.1	191	3050
[124,]	44.1	18.0	210	4000
[125,]	38.5	17.9	190	3325
[126,]	43.1	19.2	197	3500
[127,]	36.8	18.5	193	3500
[128,]	37.5	18.5	199	4475
[129,]	38.1	17.6	187	3425
[130,]	41.1	17.5	190	3900
[131,]	35.6	17.5	191	3175
[132,]	40.2	20.1	200	3975
[133,]	37.0	16.5	185	3400
[134,]	39.7	17.9	193	4250
[135,]	40.2	17.1	193	3400
[136,]	40.6	17.2	187	3475
[137,]	32.1	15.5	188	3050
[138,]	40.7	17.0	190	3725
[139,]	37.3	16.8	192	3000
[140,]	39.0	18.7	185	3650
[141,]	39.2	18.6	190	4250
[142,]	36.6	18.4	184	3475
[143,]	36.0	17.8	195	3450
[144,]	37.8	18.1	193	3750
[145,]	36.0	17.1	187	3700
[146,]	41.5	18.5	201	4000
[147,]	46.1	13.2	211	4500
[148,]	50.0	16.3	230	5700
[149,]	48.7	14.1	210	4450
[150,]	50.0	15.2	218	5700
[151,]	47.6	14.5	215	5400
[152,]	46.5	13.5	210	4550

[153,]	45.4	14.6	211	4800
[154,]	46.7	15.3	219	5200
[155,]	43.3	13.4	209	4400
[156,]	46.8	15.4	215	5150
[157,]	40.9	13.7	214	4650
[158,]	49.0	16.1	216	5550
[159,]	45.5	13.7	214	4650
[160,]	48.4	14.6	213	5850
[161,]	45.8	14.6	210	4200
[162,]	49.3	15.7	217	5850
[163,]	42.0	13.5	210	4150
[164,]	49.2	15.2	221	6300
[165,]	46.2	14.5	209	4800
[166,]	48.7	15.1	222	5350
[167,]	50.2	14.3	218	5700
[168,]	45.1	14.5	215	5000
[169,]	46.5	14.5	213	4400
[170,]	46.3	15.8	215	5050
[171,]	42.9	13.1	215	5000
[172,]	46.1	15.1	215	5100
[173,]	47.8	15.0	215	5650
[174,]	48.2	14.3	210	4600
[175,]	50.0	15.3	220	5550
[176,]	47.3	15.3	222	5250
[177,]	42.8	14.2	209	4700
[178,]	45.1	14.5	207	5050
[179,]	59.6	17.0	230	6050
[180,]	49.1	14.8	220	5150
[181,]	48.4	16.3	220	5400
[182,]	42.6	13.7	213	4950
[183,]	44.4	17.3	219	5250
[184,]	44.0	13.6	208	4350
[185,]	48.7	15.7	208	5350
[186,]	42.7	13.7	208	3950
[187,]	49.6	16.0	225	5700
[188,]	45.3	13.7	210	4300
[189,]	49.6	15.0	216	4750
[190,]	50.5	15.9	222	5550
[191,]	43.6	13.9	217	4900
[192,]	45.5	13.9	210	4200
[193,]	50.5	15.9	225	5400
[194,]	44.9	13.3	213	5100
[195,]	45.2	15.8	215	5300

[196,]	46.6	14.2	210	4850
[197,]	48.5	14.1	220	5300
[198,]	45.1	14.4	210	4400
[199,]	50.1	15.0	225	5000
[200,]	46.5	14.4	217	4900
[201,]	45.0	15.4	220	5050
[202,]	43.8	13.9	208	4300
[203,]	45.5	15.0	220	5000
[204,]	43.2	14.5	208	4450
[205,]	50.4	15.3	224	5550
[206,]	45.3	13.8	208	4200
[207,]	46.2	14.9	221	5300
[208,]	45.7	13.9	214	4400
[209,]	54.3	15.7	231	5650
[210,]	45.8	14.2	219	4700
[211,]	49.8	16.8	230	5700
[212,]	49.5	16.2	229	5800
[213,]	43.5	14.2	220	4700
[214,]	50.7	15.0	223	5550
[215,]	47.7	15.0	216	4750
[216,]	46.4	15.6	221	5000
[217,]	48.2	15.6	221	5100
[218,]	46.5	14.8	217	5200
[219,]	46.4	15.0	216	4700
[220,]	48.6	16.0	230	5800
[221,]	47.5	14.2	209	4600
[222,]	51.1	16.3	220	6000
[223,]	45.2	13.8	215	4750
[224,]	45.2	16.4	223	5950
[225,]	49.1	14.5	212	4625
[226,]	52.5	15.6	221	5450
[227,]	47.4	14.6	212	4725
[228,]	50.0	15.9	224	5350
[229,]	44.9	13.8	212	4750
[230,]	50.8	17.3	228	5600
[231,]	43.4	14.4	218	4600
[232,]	51.3	14.2	218	5300
[233,]	47.5	14.0	212	4875
[234,]	52.1	17.0	230	5550
[235,]	47.5	15.0	218	4950
[236,]	52.2	17.1	228	5400
[237,]	45.5	14.5	212	4750
[238,]	49.5	16.1	224	5650

[239,]	44.5	14.7	214	4850
[240,]	50.8	15.7	226	5200
[241,]	49.4	15.8	216	4925
[242,]	46.9	14.6	222	4875
[243,]	48.4	14.4	203	4625
[244,]	51.1	16.5	225	5250
[245,]	48.5	15.0	219	4850
[246,]	55.9	17.0	228	5600
[247,]	47.2	15.5	215	4975
[248,]	49.1	15.0	228	5500
[249,]	46.8	16.1	215	5500
[250,]	41.7	14.7	210	4700
[251,]	53.4	15.8	219	5500
[252,]	43.3	14.0	208	4575
[253,]	48.1	15.1	209	5500
[254,]	50.5	15.2	216	5000
[255,]	49.8	15.9	229	5950
[256,]	43.5	15.2	213	4650
[257,]	51.5	16.3	230	5500
[258,]	46.2	14.1	217	4375
[259,]	55.1	16.0	230	5850
[260,]	48.8	16.2	222	6000
[261,]	47.2	13.7	214	4925
[262,]	46.8	14.3	215	4850
[263,]	50.4	15.7	222	5750
[264,]	45.2	14.8	212	5200
[265,]	49.9	16.1	213	5400
[266,]	46.5	17.9	192	3500
[267,]	50.0	19.5	196	3900
[268,]	51.3	19.2	193	3650
[269,]	45.4	18.7	188	3525
[270,]	52.7	19.8	197	3725
[271,]	45.2	17.8	198	3950
[272,]	46.1	18.2	178	3250
[273,]	51.3	18.2	197	3750
[274,]	46.0	18.9	195	4150
[275,]	51.3	19.9	198	3700
[276,]	46.6	17.8	193	3800
[277,]	51.7	20.3	194	3775
[278,]	47.0	17.3	185	3700
[279,]	52.0	18.1	201	4050
[280,]	45.9	17.1	190	3575
[281,]	50.5	19.6	201	4050

[282,]	50.3	20.0	197	3300
[283,]	58.0	17.8	181	3700
[284,]	46.4	18.6	190	3450
[285,]	49.2	18.2	195	4400
[286,]	42.4	17.3	181	3600
[287,]	48.5	17.5	191	3400
[288,]	43.2	16.6	187	2900
[289,]	50.6	19.4	193	3800
[290,]	46.7	17.9	195	3300
[291,]	52.0	19.0	197	4150
[292,]	50.5	18.4	200	3400
[293,]	49.5	19.0	200	3800
[294,]	46.4	17.8	191	3700
[295,]	52.8	20.0	205	4550
[296,]	40.9	16.6	187	3200
[297,]	54.2	20.8	201	4300
[298,]	42.5	16.7	187	3350
[299,]	51.0	18.8	203	4100
[300,]	49.7	18.6	195	3600
[301,]	47.5	16.8	199	3900
[302,]	47.6	18.3	195	3850
[303,]	52.0	20.7	210	4800
[304,]	46.9	16.6	192	2700
[305,]	53.5	19.9	205	4500
[306,]	49.0	19.5	210	3950
[307,]	46.2	17.5	187	3650
[308,]	50.9	19.1	196	3550
[309,]	45.5	17.0	196	3500
[310,]	50.9	17.9	196	3675
[311,]	50.8	18.5	201	4450
[312,]	50.1	17.9	190	3400
[313,]	49.0	19.6	212	4300
[314,]	51.5	18.7	187	3250
[315,]	49.8	17.3	198	3675
[316,]	48.1	16.4	199	3325
[317,]	51.4	19.0	201	3950
[318,]	45.7	17.3	193	3600
[319,]	50.7	19.7	203	4050
[320,]	42.5	17.3	187	3350
[321,]	52.2	18.8	197	3450
[322,]	45.2	16.6	191	3250
[323,]	49.3	19.9	203	4050
[324,]	50.2	18.8	202	3800

[325,]	45.6	19.4	194	3525
[326,]	51.9	19.5	206	3950
[327,]	46.8	16.5	189	3650
[328,]	45.7	17.0	195	3650
[329,]	55.8	19.8	207	4000
[330,]	43.5	18.1	202	3400
[331,]	49.6	18.2	193	3775
[332,]	50.8	19.0	210	4100
[333,]	50.2	18.7	198	3775

Originally tried creating the matrix using just “`matrix(penguins_complete) ...`” but this doesn’t work because it treats each column as one big element. Therefore, I needed to use `as.matrix` because this is what’s able to handle the dataframe properly and return all the elements of each column instead of simply the column name.

I also left the space before the comma in the `penguin_measurements` line blank because each matrix is created in terms of [rows, columns]. I wanted each column to represent the elements in a variable so I left the row area blank and this allows each row to have the element’s index.

```
# 3.2: Apply functions
apply(penguin_measurements, 2, mean) # Returns the mean of each column
```

bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
43.99279	17.16486	200.96697	4207.05706

```
apply(penguin_measurements, 2, which.max)
```

bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
179	15	209	164

```
# which.max function returns the index of the maximum value per column.
```

```
# 4.1 Write standardize function
```

```
standardize <- function(x) {  
  x <- x[!is.na(x)]  
  (x - mean(x)) / sd(x)  
}
```

```
bill_length_standardized <- standardize(penguins$bill_length_mm)  
mean(bill_length_standardized)
```

```
[1] -6.882084e-17
```

```
sd(bill_length_standardized)
```

```
[1] 1
```

explaining function creation: This is a logical vector and will return only the values that are TRUE. In this case, the ! makes the non-NA values TRUE (flips the definition) so it will only return the non-NA values. I saved the vector to a variable so that I can perform operations on the whole vector to return $(x - \text{mean}) / \text{sd}$.

```
# Bonus: Option A
```

```
adelie <- penguins[penguins$species == "Adelie", ]  
chinstrap <- penguins[penguins$species == "Chinstrap", ]  
gentoo <- penguins[penguins$species == "Gentoo", ]  
# created variables storing the different penguin species.
```

```
species_summary <- data.frame(  
  species = c("Adelie", "Chinstrap", "Gentoo"),  
  # concatenated the different species in a column  
  count = c(nrow(adelie), nrow(chinstrap), nrow(gentoo)),  
  # using nrow returns the number of elements in a row  
  mean_body_mass_g = c(mean(adelie$body_mass_g, na.rm = TRUE),  
  # calculated the mean of each different species of penguin separately  
  # remembering to ignore missing values using "na.rm"
```

```

        mean(chinstrap$body_mass_g, na.rm = TRUE),
        mean(gentoo$body_mass_g, na.rm = TRUE)),
mean_flipper_length_mm = c(mean(adelie$flipper_length_mm, na.rm = TRUE),
# calculated the mean of each species' flipper length
# also ignored missing values using "na.rm"
        mean(chinstrap$flipper_length_mm, na.rm = TRUE),
        mean(gentoo$flipper_length_mm, na.rm = TRUE))
)
species_summary

```

	species	count	mean_body_mass_g	mean_flipper_length_mm
1	Adelie	152	3700.662	189.9536
2	Chinstrap	68	3733.088	195.8235
3	Gentoo	124	5076.016	217.1870